KCE PATHWAYS · GRADES 4–5 · UNIT 4: SHAPES AND SPACE · WEEK 17 · MATH

Market Morning in Honolulu — Teacher Lesson Plan & Answer Key

Learning objectives: Students divide decimals by whole numbers (placing the decimal point straight above, writing placeholder zeros), divide by 10, 100 and 1,000 using place-value patterns, find unit prices and convert metric measurements, estimate first to check where the decimal point belongs, check by multiplying, and solve two-step and work-backward problems.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Slide 0 — Intro and name: meet Leilani, who helps her grandmother Tutu Mahina at a Saturday farmers market in Kakaʻako, Honolulu, with her cousin Caleb and Mr. Haddad at the juice tent. The student types first and last name so results are sent. Read the scoring rules together: 2 tries per question, Pathway Points only for a correct first try, and the answer plus the reason appear after the 2nd miss. Typed number answers ignore capital letters, commas in large numbers, dollar signs and a final period.
2. Slide 1 — Warm-Up, Part 1: Dividing with Unit Fractions (4 pts, 2 each, Week 16 skill).
3. Slide 2 — Warm-Up, Part 2 (4 pts, 2 each): one multiple-choice and one typed item.
4. Slide 3 — Pencil Break (Pre): the student opens or prints the Pre/Post sheet (grade4-5/prepost/KCE_Math_PrePost_G4-5W17.pdf) and completes Page 1 (Pre-Lesson Warm-Up) off screen before any teaching.
5. Slide 4 — Fruit from the Islands: nonfiction connection about a Hawaiʻi farmers market. No questions.
6. Slide 5 — Picture It: A Decimal Divided by a Whole Number. Mr. Haddad's 4.8 liters of juice in 4 pitchers is 1.2 liters each. Stress: equal shares, same as always; line the decimal point up straight above.
7. Slide 6 — Your Turn, Part 1: Decimals ÷ Whole Numbers (8 pts, 4 each).
8. Slide 7 — Your Turn, Part 2 (4 pts): one more, watching for the placeholder zero (0.42 ÷ 7 = 0.06).
9. Slide 8 — Teach: Dividing by 10, 100, and 1,000. Caleb's 456 mL tasting cups: each divide by 10 moves every digit one place to the right.
10. Slide 9 — Your Turn, Part 1: Dividing by 10, 100, and 1,000 (8 pts, 4 each).
11. Slide 10 — Your Turn, Part 2 (4 pts): find the missing divisor (10, 100 or 1,000).
12. Slide 11 — Teach: Money and Measurement. Tutu Mahina's fruit stand: price of one item, and metric conversions.
13. Slide 12 — Your Turn, Part 1: Money and Measurement (8 pts, 4 each).
14. Slide 13 — Your Turn, Part 2 (4 pts): one more problem.
15. Slide 14 — Vocabulary Round-Up: 8 flip cards (dividend, divisor, quotient, decimal point, placeholder zero, place value, unit price, estimate).
16. Slide 15 — Your Turn: Using the Words (6 pts, 3 each): one multiple-choice and one typed vocabulary item.
17. Slide 16 — Teach: Multi-Step Problems and Checking Your Work. Two divisions in a row (pitchers, then cups) and working backward with multiplication.
18. Slide 17 — Your Turn: Two Steps and Working Backward (6 pts, 3 each).

19. Slide 18 — Real World: The Fruit Stand (4 pts): a different money or coconut-water situation in each set.
20. Slide 19 — Real World: Measuring at the Market (4 pts): kilograms, kilometers or liters shared equally.
21. Slide 20 — Think It Through: Explain and Find the Mistake (10 pts, 5 each): a “which explanation is correct?” item and a “find the mistake” item. Ask the student to say why each wrong choice is wrong.
22. Slide 21 — Fix the Sentence, Part 1 (4 pts, 2 each): typed sentences checked exactly (capital letters, quotation marks, commas, end marks, and one grammar fix). Straight or curly quotes and extra spaces are OK. The corrected sentence appears after the 2nd miss.
23. Slide 22 — Fix the Sentence, Part 2 (4 pts, 2 each): two more sentences, same rules. Where shown, both serial-comma and no-serial-comma forms of a list are accepted, and one item also accepts either verb tense.
24. Slide 23 — Match-Up Round (10 pts): 5 problems matched to answers (decimal quotients, money, and metric conversions), 2 pts each on the first try.
25. Slide 24 — Speed Trail (bonus XP only): 3 quick items against the student’s own best time.
26. Slide 25 — Looking Back reflection (bonus XP only).
27. Slide 26 — Boss Battle: Closing Time at the Market (8 pts, 4 each): an end-of-day situation in each set, then multi-step questions (for example, liters in each bottle, or how much more one mango costs in a small bag than in a big crate).
28. Slide 27 — Pencil Break (Post): the student completes Page 2 (Post-Lesson Challenge) of the same sheet (grade4-5/prepost/KCE_Math_PrePost_G4-5W17.pdf), plus the ★ Challenge if time allows, before seeing the score.
29. Slide 28 — Results (Market Morning Complete!): the score out of 100 is sent to KingCretot Experience. 85 or higher shows “Market Decimal Master!” with the congratulations message “You helped Tutu Mahina’s fruit stand have a great Saturday in Honolulu!” and a big confetti burst; below 85 shows “Good Work at the Market!” with encouragement (“You are getting closer every time.”), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Post-Lesson Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: DIVIDING WITH UNIT FRACTIONS (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
r1_s0	Set 1: A unit fraction ÷ a whole number (Week 16): type the quotient as a fraction. 1/6 ÷ 4 =	**1/24**	Splitting 1/6 into 4 equal parts makes twenty-fourths, because 6 × 4 = 24. So 1/6 ÷ 4 = 1/24. Check: 1/24 × 4 = 4/24 = 1/6.
r2_s0	Set 1: A whole number ÷ a unit fraction (Week 16): type the quotient. 9 ÷ 1/3 =	**27** (also accepted: 27/1)	Each whole holds 3 thirds, so 9 wholes hold 9 × 3 = 27 thirds. 9 ÷ 1/3 = 27.
r3_s0	Set 1: Which explanation is correct for 1/4 ÷ 4? [1/16, because splitting one fourth into 4 equal parts makes sixteenths / 1, because 4 fourths make 1 whole / 1/8, because 4 + 4 = 8 / 16, because 4 × 4 = 16]	**1/16, because splitting one fourth into 4 equal parts makes sixteenths**	Splitting one fourth into 4 equal parts makes 16 equal parts in the whole, so each part is 1/16. Check: 1/16 × 4 = 4/16 = 1/4.
r4_s0	Set 1: Leilani pours 3 liters of pineapple juice into small tasting cups that each hold 1/5 liter. How many cups does she fill? Cups:	**15** (also accepted: 15 cups, 15 cup)	Each liter fills 5 cups of 1/5 liter, so 3 liters fill 3 × 5 = 15 cups. 3 ÷ 1/5 = 15.

ID	Question	Answer	Why (explain to student)
r1_s1	Set 2: A unit fraction ÷ a whole number (Week 16): type the quotient as a fraction. 1/7 ÷ 3 =	**1/21**	Splitting 1/7 into 3 equal parts makes twenty-firsts, because 7 × 3 = 21. So 1/7 ÷ 3 = 1/21. Check: 1/21 × 3 = 3/21 = 1/7.
r2_s1	Set 2: A whole number ÷ a unit fraction (Week 16): type the quotient. 4 ÷ 1/6 =	**24** (also accepted: 24/1)	Each whole holds 6 sixths, so 4 wholes hold 4 × 6 = 24 sixths. 4 ÷ 1/6 = 24.
r3_s1	Set 2: Which explanation is correct for 6 ÷ 1/3? [18, because each of the 6 wholes holds 3 thirds / 2, because 6 ÷ 3 = 2 / 1/18, because 3 × 6 = 18 / 6/3, because you put 6 over 3]	**18, because each of the 6 wholes holds 3 thirds**	Each of the 6 wholes holds 3 thirds, so 6 ÷ 1/3 = 6 × 3 = 18. Dividing by a unit fraction gives MORE pieces, not fewer.
r4_s1	Set 2: Tutu Mahina shares 1/4 pound of macadamia nuts equally among 3 neighbors. What fraction of a pound does each neighbor get? Pound:	**1/12** (also accepted: 1/12 pound, 1/12 pounds, 1/12 lb)	Split 1/4 pound into 3 equal shares: 1/4 ÷ 3 = 1/12 pound, because 4 × 3 = 12.
r1_s2	Set 3: A unit fraction ÷ a whole number (Week 16): type the quotient as a fraction. 1/9 ÷ 2 =	**1/18**	Splitting 1/9 into 2 equal parts makes eighteenths, because 9 × 2 = 18. So 1/9 ÷ 2 = 1/18. Check: 1/18 × 2 = 2/18 = 1/9.
r2_s2	Set 3: A whole number ÷ a unit fraction (Week 16): type the quotient. 7 ÷ 1/5 =	**35** (also accepted: 35/1)	Each whole holds 5 fifths, so 7 wholes hold 7 × 5 = 35 fifths. 7 ÷ 1/5 = 35.
r3_s2	Set 3: Which explanation is correct for 1/6 ÷ 2? [1/12, because splitting one sixth into 2 equal parts makes twelfths / 3, because 6 ÷ 2 = 3 / 1/3, because 6 ÷ 2 = 3, so the answer is 1/3 / 12, because 6 × 2 = 12]	**1/12, because splitting one sixth into 2 equal parts makes twelfths**	Splitting one sixth into 2 equal parts makes 12 equal parts in the whole, so each part is 1/12. Check: 1/12 × 2 = 2/12 = 1/6.
r4_s2	Set 3: After the market, Caleb plans a 2-mile walk with Tutu Mahina. He marks a stop on his map every 1/8 mile. How many 1/8-mile sections are there? Sections:	**16** (also accepted: 16 sections, 16 section)	Each mile has 8 sections of 1/8 mile, so 2 miles have 2 × 8 = 16 sections. 2 ÷ 1/8 = 16.

YOUR TURN: DECIMALS ÷ WHOLE NUMBERS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
l1_s0	Set 1: Mr. Haddad pours 9.6 liters of juice equally into 4 pitchers. How many liters go in each pitcher? Liters:	**2.4** (also accepted: 2.40, 2.4 liters, 2.4 liter, 2.4 L)	9 ones ÷ 4 = 2 ones with 1 one left over. That 1 one is 10 tenths, plus 6 tenths makes 16 tenths, and 16 tenths ÷ 4 = 4 tenths. So 9.6 ÷ 4 = 2.4 liters. Check: 2.4 × 4 = 9.6.
l2_s0	Set 1: Write a zero to keep dividing. Type the exact quotient. 6.3 ÷ 2 =	**3.15**	6.3 ÷ 2 = 3.1 with 1 tenth left over. Write 6.3 as 6.30 and keep dividing: 10 hundredths ÷ 2 = 5 hundredths. So 6.3 ÷ 2 = 3.15. Check: 3.15 × 2 = 6.30.
l3_s0	Set 1: Watch for the placeholder zero. Type the quotient. 0.42 ÷ 7 =	**0.06**	0.42 is 42 hundredths. 42 hundredths ÷ 7 = 6 hundredths = 0.06. The 0 in the tenths place is a placeholder. (0.6 would be 6 tenths, which is ten times too big.)

ID	Question	Answer	Why (explain to student)
l1_s1	Set 2: Tutu Mahina packs 7.5 kilograms of papayas equally into 3 baskets. How many kilograms go in each basket? Kilograms:	**2.5** (also accepted: 2.50, 2.5 kilograms, 2.5 kilogram, 2.5 kg)	7 ones ÷ 3 = 2 ones with 1 one left over. That 1 one is 10 tenths, plus 5 tenths makes 15 tenths, and 15 tenths ÷ 3 = 5 tenths. So 7.5 ÷ 3 = 2.5 kilograms. Check: 2.5 × 3 = 7.5.
l2_s1	Set 2: Write a zero to keep dividing. Type the exact quotient. 4.6 ÷ 4 =	**1.15**	4.6 ÷ 4 = 1.1 with 2 tenths left over. Write 4.6 as 4.60 and keep dividing: 20 hundredths ÷ 4 = 5 hundredths. So 4.6 ÷ 4 = 1.15. Check: 1.15 × 4 = 4.60.
l3_s1	Set 2: Watch for the placeholder zero. Type the quotient. 0.35 ÷ 5 =	**0.07**	0.35 is 35 hundredths. 35 hundredths ÷ 5 = 7 hundredths = 0.07. The 0 in the tenths place is a placeholder. (0.7 would be 7 tenths, which is ten times too big.)
l1_s2	Set 3: Caleb bikes 6.6 kilometers delivering fruit boxes to neighbors. He bikes the same distance on each of 3 days. How many kilometers does he bike each day? Kilometers:	**2.2** (also accepted: 2.20, 2.2 kilometers, 2.2 kilometer, 2.2 km)	6 ones ÷ 3 = 2 ones, and 6 tenths ÷ 3 = 2 tenths. So 6.6 ÷ 3 = 2.2 kilometers. Check: 2.2 × 3 = 6.6.
l2_s2	Set 3: Write a zero to keep dividing. Type the exact quotient. 9.9 ÷ 6 =	**1.65**	9.9 ÷ 6 = 1.6 with 3 tenths left over. Write 9.9 as 9.90 and keep dividing: 30 hundredths ÷ 6 = 5 hundredths. So 9.9 ÷ 6 = 1.65. Check: 1.65 × 6 = 9.90.
l3_s2	Set 3: Watch for the placeholder zero. Type the quotient. 0.32 ÷ 8 =	**0.04**	0.32 is 32 hundredths. 32 hundredths ÷ 8 = 4 hundredths = 0.04. The 0 in the tenths place is a placeholder. (0.4 would be 4 tenths, which is ten times too big.)

YOUR TURN: DIVIDING BY 10, 100, AND 1,000 (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
k1_s0	Set 1: Type the quotient. 56.2 ÷ 10 =	**5.62**	Dividing by 10 moves every digit one place to the right: the 5 moves from tens to ones, the 6 from ones to tenths, and the 2 from tenths to hundredths. 56.2 ÷ 10 = 5.62.
k2_s0	Set 1: Type the quotient. Fill any empty places with zeros. 3.8 ÷ 100 =	**0.038**	Dividing by 100 moves every digit two places to the right: the 3 moves from ones to hundredths and the 8 from tenths to thousandths. A 0 holds the tenths place. 3.8 ÷ 100 = 0.038.
k3_s0	Set 1: Find the missing divisor: 10, 100, or 1,000? 71.3 ÷ ___ = 0.713	**100**	Each digit moved two places to the right (the 7 went from tens to tenths), so the divisor is 100. 71.3 ÷ 100 = 0.713.

ID	Question	Answer	Why (explain to student)
k1_s1	Set 2: Type the quotient. 409 ÷ 100 =	**4.09**	Dividing by 100 moves every digit two places to the right: the 4 moves from hundreds to ones, the 0 from tens to tenths, and the 9 from ones to hundredths. 409 ÷ 100 = 4.09.
k2_s1	Set 2: Type the quotient. Fill any empty places with zeros. 65 ÷ 1,000 =	**0.065**	Dividing by 1,000 moves every digit three places to the right: the 6 moves from tens to hundredths and the 5 from ones to thousandths. A 0 holds the tenths place. 65 ÷ 1,000 = 0.065.
k3_s1	Set 2: Find the missing divisor: 10, 100, or 1,000? 28.4 ÷ ___ = 2.84	**10**	Each digit moved one place to the right (the 2 went from tens to ones), so the divisor is 10. 28.4 ÷ 10 = 2.84.
k1_s2	Set 3: Type the quotient. 93.7 ÷ 10 =	**9.37**	Dividing by 10 moves every digit one place to the right: the 9 moves from tens to ones, the 3 from ones to tenths, and the 7 from tenths to hundredths. 93.7 ÷ 10 = 9.37.
k2_s2	Set 3: Type the quotient. Fill any empty places with zeros. 0.9 ÷ 100 =	**0.009**	Dividing by 100 moves every digit two places to the right: the 9 moves from tenths to thousandths. Zeros hold the tenths and hundredths places. 0.9 ÷ 100 = 0.009.
k3_s2	Set 3: Find the missing divisor: 10, 100, or 1,000? 1,520 ÷ ___ = 1.52	**1,000** (also accepted: 1000)	Each digit moved three places to the right (the 1 went from thousands to ones), so the divisor is 1,000. 1,520 ÷ 1,000 = 1.52.

YOUR TURN: MONEY AND MEASUREMENT (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
e1_s0	Set 1: A bag of 6 mangoes costs $8.70. What is the unit price, the cost of one mango? Dollars: $	**1.45** (also accepted: $1.45, 1.45 dollars)	8 ÷ 6 = 1 with 2 left over. 2 ones and 7 tenths make 27 tenths, and 27 tenths ÷ 6 = 4 tenths with 3 tenths left. 3 tenths is 30 hundredths, and 30 hundredths ÷ 6 = 5 hundredths. $8.70 ÷ 6 = $1.45. Check: $1.45 × 6 = $8.70.
e2_s0	Set 1: The banner over Tutu Mahina's tent is 345 centimeters long. How many meters long is it? Meters:	**3.45** (also accepted: 3.45 meters, 3.45 meter, 3.45 m)	1 meter = 100 centimeters. Centimeters are smaller, so divide by 100: 345 ÷ 100 = 3.45 meters.
e3_s0	Set 1: A container of poke has a mass of 850 grams. What is its mass in kilograms? Kilograms:	**0.85** (also accepted: 0.85 kilograms, 0.85 kilogram, 0.85 kg)	1 kilogram = 1,000 grams. Grams are smaller, so divide by 1,000: 850 ÷ 1,000 = 0.85 kilogram. That is less than 1 kilogram, which makes sense because 850 grams is less than 1,000 grams.
e1_s1	Set 2: Leilani, Caleb, Tutu Mahina, and Mr. Haddad buy four cups of shave ice for $15.20 and split the cost equally. How much does each person pay? Dollars: $	**3.80** (also accepted: $3.80, 3.80 dollars, 3.8 dollars)	15 ÷ 4 = 3 with 3 left over. 3 ones and 2 tenths make 32 tenths, and 32 tenths ÷ 4 = 8 tenths. $15.20 ÷ 4 = $3.80. Check: $3.80 × 4 = $15.20.

ID	Question	Answer	Why (explain to student)
e2_s1	Set 2: A glass jug holds 1,250 milliliters of coconut water. How many liters is that? Liters:	**1.25** (also accepted: 1.25 liters, 1.25 liter, 1.25 L)	1 liter = 1,000 milliliters. Milliliters are smaller, so divide by 1,000: 1,250 ÷ 1,000 = 1.25 liters.
e3_s1	Set 2: The bamboo pole that holds up the price sign is 62 centimeters long. How many meters long is it? Meters:	**0.62** (also accepted: 0.62 meters, 0.62 meter, 0.62 m)	1 meter = 100 centimeters. Centimeters are smaller, so divide by 100: 62 ÷ 100 = 0.62 meter. That is less than 1 meter, which makes sense.
e1_s2	Set 3: Three big bags of macadamia nuts cost $41.85 together. Each bag costs the same. What is the price of one bag? Dollars: $	**13.95** (also accepted: $13.95, 13.95 dollars)	41 ÷ 3 = 13 with 2 left over. 2 ones and 8 tenths make 28 tenths, and 28 tenths ÷ 3 = 9 tenths with 1 tenth left. 1 tenth is 10 hundredths, plus 5 hundredths makes 15 hundredths, and 15 ÷ 3 = 5. $41.85 ÷ 3 = $13.95. Check: $13.95 × 3 = $41.85.
e2_s2	Set 3: Mr. Haddad's juice cooler holds 4,500 milliliters. How many liters is that? Liters:	**4.5** (also accepted: 4.50, 4.5 liters, 4.5 liter, 4.5 L)	1 liter = 1,000 milliliters. Milliliters are smaller, so divide by 1,000: 4,500 ÷ 1,000 = 4.5 liters.
e3_s2	Set 3: A bag of dried mango has a mass of 375 grams. What is its mass in kilograms? Kilograms:	**0.375** (also accepted: 0.375 kilograms, 0.375 kilogram, 0.375 kg)	1 kilogram = 1,000 grams. Grams are smaller, so divide by 1,000: 375 ÷ 1,000 = 0.375 kilogram.

YOUR TURN: USING THE WORDS (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
v1_s0	Set 1: Read the sentence from the juice tent: "Mr. Haddad poured 9.6 liters into 4 pitchers, so each pitcher held 2.4 liters." In this division, 2.4 is the ___. [Quotient / Dividend / Divisor / Placeholder zero]	**Quotient**	The quotient is the answer to a division problem. 9.6 ÷ 4 = 2.4, so 2.4 is the quotient. 9.6 is the dividend and 4 is the divisor.
v2_s0	Set 1: Type the word: "Tutu Mahina packed 7.5 kilograms of papayas equally into 3 baskets." In 7.5 ÷ 3, the number 3 is called the ___.	**divisor** (also accepted: the divisor, a divisor)	The divisor is the number you divide by. In 7.5 ÷ 3, you divide by 3 (the number of baskets), so 3 is the divisor.
v1_s1	Set 2: Read the sentence: "Leilani wrote 0.35 ÷ 5 = 0.07, with a 0 in the tenths place so the 7 stays in the hundredths place." That 0 is called a ___. [Placeholder zero / Quotient / Divisor / Unit price]	**Placeholder zero**	A placeholder zero holds an empty place. In 0.07, the 0 holds the tenths place so the 7 stays in the hundredths place.
v2_s1	Set 2: Type the word: "Mr. Haddad poured 9.6 liters of juice equally into 4 pitchers." In 9.6 ÷ 4, the amount being divided, 9.6, is called the ___.	**dividend** (also accepted: the dividend, a dividend)	The dividend is the amount being divided. Mr. Haddad is splitting up 9.6 liters, so 9.6 is the dividend.
v1_s2	Set 3: Read the sentence from the fruit stand: "A box of 4 pineapples costs $31.00, so one pineapple costs $7.75." The cost of ONE pineapple is called the ___. [Unit price / Dividend / Placeholder zero / Divisor]	**Unit price**	The unit price is the cost of one item. $31.00 ÷ 4 = $7.75 for one pineapple.
v2_s2	Set 3: Type the word: "Before dividing 7.56 by 3, Leilani said the answer should be between 2 and 3." A close guess like this, made to check an answer, is called an ___.	**estimate** (also accepted: an estimate, estimation)	An estimate is a close guess that helps you check your work. 6 ÷ 3 = 2 and 9 ÷ 3 = 3, so 7.56 ÷ 3 must be between 2 and 3.

YOUR TURN: TWO STEPS AND WORKING BACKWARD (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
t1_s0	Set 1: Two steps: 19.2 liters of juice are poured equally into 8 pitchers. Each pitcher is then poured equally into 10 cups. How many liters are in each cup? Liters:	**0.24** (also accepted: 0.24 liters, 0.24 liter, 0.24 L)	Step 1: 19.2 ÷ 8 = 2.4 liters in each pitcher. Step 2: 2.4 ÷ 10 = 0.24 liter in each cup. Check: 0.24 × 10 = 2.4 and 2.4 × 8 = 19.2.
t2_s0	Set 1: Work backward: find the missing dividend. ___ ÷ 8 = 0.45	**3.6** (also accepted: 3.60)	Work backward with multiplication: the dividend = quotient × divisor = 0.45 × 8 = 3.6. Check: 3.6 ÷ 8 = 0.45.
t1_s1	Set 2: Two steps: Caleb cuts 14.4 meters of ribbon into 4 equal long pieces. Then he cuts each long piece into 10 equal short pieces. How many meters long is each short piece? Meters:	**0.36** (also accepted: 0.36 meters, 0.36 meter, 0.36 m)	Step 1: 14.4 ÷ 4 = 3.6 meters for each long piece. Step 2: 3.6 ÷ 10 = 0.36 meter for each short piece. Check: 0.36 × 10 = 3.6 and 3.6 × 4 = 14.4.
t2_s1	Set 2: Work backward: find the missing dividend. ___ ÷ 6 = 0.35	**2.1** (also accepted: 2.10)	Work backward with multiplication: the dividend = quotient × divisor = 0.35 × 6 = 2.1. Check: 2.1 ÷ 6 = 0.35.
t1_s2	Set 3: Two steps: 7.2 kilograms of poke are split equally onto 3 trays. Each tray is split into 100 equal tasting samples. What is the mass of one sample in kilograms? Kilograms:	**0.024** (also accepted: 0.024 kilograms, 0.024 kilogram, 0.024 kg)	Step 1: 7.2 ÷ 3 = 2.4 kilograms on each tray. Step 2: 2.4 ÷ 100 = 0.024 kilogram for each sample (every digit moves two places to the right). Check: 0.024 × 100 = 2.4 and 2.4 × 3 = 7.2.
t2_s2	Set 3: Work backward: find the missing dividend. ___ ÷ 4 = 1.15	**4.6** (also accepted: 4.60)	Work backward with multiplication: the dividend = quotient × divisor = 1.15 × 4 = 4.6. Check: 4.6 ÷ 4 = 1.15.

REAL WORLD: THE FRUIT STAND (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: *The Situation Tutu Mahina sells a gift box of 4 jars of liliko'i (passion fruit) butter for $38.00. A single jar bought on its own costs $10.25.* Two steps: how much money do you save on EACH jar by buying the gift box instead of single jars? Dollars: $	**0.75** (also accepted: $0.75, 0.75 dollars, 75 cents)	Step 1: $38.00 ÷ 4 = $9.50 per jar in the box. Step 2: $10.25 − $9.50 = $0.75 saved on each jar.
w1_s1	Set 2: *The Situation The juice tent has 3 big coolers that hold 182.4 liters of coconut water in all, with the same amount in each cooler. Caleb empties one cooler equally into 10 jugs to carry to the serving table.* Two steps: how many liters of coconut water go in each jug? Liters:	**6.08** (also accepted: 6.08 liters, 6.08 liter, 6.08 L)	Step 1: 182.4 ÷ 3 = 60.8 liters in one cooler. Step 2: 60.8 ÷ 10 = 6.08 liters in each jug. Check: 6.08 × 10 = 60.8 and 60.8 × 3 = 182.4.
w1_s2	Set 3: *The Situation Tutu Mahina buys 5 packs of paper fruit bags for $13.75. Each pack costs the same. Next week she wants to buy 8 more packs at the same price.* Two steps: how much will the 8 new packs cost? Dollars: $	**22** (also accepted: 22.00, $22, $22.00, 22 dollars)	Step 1: $13.75 ÷ 5 = $2.75 for one pack. Step 2: $2.75 × 8 = $22.00 for 8 packs.

REAL WORLD: MEASURING AT THE MARKET (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w2_s0	Set 1: *The Situation Leilani is baking macadamia nut cookies for the fruit stand. Her recipe uses 0.6 kilogram of macadamia nuts for 4 equal batches.* Two steps: how many GRAMS of macadamia nuts go into one batch? Grams:	**150** (also accepted: 150 grams, 150 gram, 150 g)	Step 1: 0.6 ÷ 4 = 0.15 kilogram per batch (6 tenths is 60 hundredths, and 60 ÷ 4 = 15). Step 2: 1 kilogram = 1,000 grams, so 0.15 × 1,000 = 150 grams.
w2_s1	Set 2: *The Situation After the market closes, helpers pick up litter along a 1.5-kilometer route nearby. Tutu Mahina divides the route into 6 equal sections, one for each helper.* Two steps: how many METERS long is each section? Meters:	**250** (also accepted: 250 meters, 250 meter, 250 m)	Step 1: 1.5 ÷ 6 = 0.25 kilometer per section (15 tenths ÷ 6 = 2 tenths with 3 tenths left; 30 hundredths ÷ 6 = 5 hundredths). Step 2: 1 kilometer = 1,000 meters, so 0.25 × 1,000 = 250 meters.
w2_s2	Set 3: *The Situation Caleb has a pitcher with 2.4 liters of guava juice. He pours it equally into 8 cups.* Two steps: how many MILLILITERS of juice go in each cup? Milliliters:	**300** (also accepted: 300 milliliters, 300 milliliter, 300 mL)	Step 1: 2.4 ÷ 8 = 0.3 liter per cup (24 tenths ÷ 8 = 3 tenths). Step 2: 1 liter = 1,000 milliliters, so 0.3 × 1,000 = 300 milliliters.

THINK IT THROUGH: EXPLAIN AND FIND THE MISTAKE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
x1_s0	Set 1: Which explanation is correct for 6.3 ÷ 3? [2.1, because 6 ones ÷ 3 = 2 ones and 3 tenths ÷ 3 = 1 tenth / 21, because 63 ÷ 3 = 21 / 0.21, because dividing always moves the decimal point two places / 2.01, because 3 ÷ 3 = 1 goes in the hundredths place]	**2.1, because 6 ones ÷ 3 = 2 ones and 3 tenths ÷ 3 = 1 tenth**	6.3 is 6 ones and 3 tenths. 6 ones ÷ 3 = 2 ones and 3 tenths ÷ 3 = 1 tenth, so 6.3 ÷ 3 = 2.1. Check: 2.1 × 3 = 6.3.
x2_s0	Set 1: Find the mistake: Leilani says 0.36 ÷ 4 = 0.9. What went wrong? [She left out the placeholder zero. 36 hundredths ÷ 4 = 9 hundredths, so the answer is 0.09 / She should multiply instead: 0.36 × 4 = 1.44 / The answer should be 9, because 36 ÷ 4 = 9 / She is right. 0.36 ÷ 4 = 0.9]	**She left out the placeholder zero. 36 hundredths ÷ 4 = 9 hundredths, so the answer is 0.09**	0.36 is 36 hundredths, and 36 hundredths ÷ 4 = 9 hundredths = 0.09. Checking her answer shows the mistake: 0.9 × 4 = 3.6, not 0.36.
x1_s1	Set 2: Which explanation is correct for 45.7 ÷ 100? [0.457, because dividing by 100 moves every digit two places to the right / 4,570, because 100 has two zeros, so you add two zeros / 4.57, because dividing by 100 moves every digit one place / 45.700, because you write two zeros after the last digit]	**0.457, because dividing by 100 moves every digit two places to the right**	Dividing by 100 moves every digit two places to the right: the 4 moves from tens to tenths, the 5 from ones to hundredths, and the 7 from tenths to thousandths. 45.7 ÷ 100 = 0.457.
x2_s1	Set 2: Find the mistake: Caleb says 5.4 ÷ 4 = 1.3 remainder 2, and he stops. What should he do? [Write 5.4 as 5.40 and keep dividing. The answer is 1.35 / Put the remainder after the decimal. The answer is 1.32 / Drop the remainder. The answer is 1.3 / Nothing. A remainder is the best answer for decimals]	**Write 5.4 as 5.40 and keep dividing. The answer is 1.35**	With decimals you can keep dividing. 5.4 = 5.40, and the 2 tenths left over are 20 hundredths: 20 ÷ 4 = 5 hundredths. So 5.4 ÷ 4 = 1.35. Check: 1.35 × 4 = 5.40.

ID	Question	Answer	Why (explain to student)
x1_s2	Set 3: Which explanation is correct? What is a sensible estimate for $23.88 ÷ 4? [About $6, because $23.88 is close to $24 and $24 ÷ 4 = $6 / About $60, because 240 ÷ 4 = 60 / About $0.60, because dividing a decimal always gives a tiny number / About $20, because $23.88 − 4 is about $20]	**About $6, because $23.88 is close to $24 and $24 ÷ 4 = $6**	$23.88 is close to $24, and $24 ÷ 4 = $6, so the answer is about $6. The exact answer is $5.97, which is very close to the estimate.
x2_s2	Set 3: Find the mistake: Mr. Haddad's helper says 7.56 ÷ 3 = 25.2. What went wrong? [The decimal point is in the wrong place. It goes straight above the one in the dividend, so the answer is 2.52 / The answer should be 252, because 756 ÷ 3 = 252 / The answer should be 0.252, because you move the decimal point three places / Nothing is wrong, because 25.2 × 3 = 7.56]	**The decimal point is in the wrong place. It goes straight above the one in the dividend, so the answer is 2.52**	The decimal point in the quotient goes straight above the decimal point in the dividend: 7.56 ÷ 3 = 2.52. An estimate catches the mistake: 7.56 is less than 9, and 9 ÷ 3 = 3, so the answer must be less than 3. Also, 25.2 × 3 = 75.6, not 7.56.

FIX THE SENTENCE (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
s1_s0	Set 1: Fix this sentence: leilani asked, how many papayas fit in one basket	**Leilani asked, "How many papayas fit in one basket?"**	Capitalize the name Leilani, put quotation marks around her exact words, capitalize the first word of the quote, and put the question mark inside the closing quotation mark.
s2_s0	Set 1: Fix this sentence: The fruit stand sells it's papayas in paper bags.	**The fruit stand sells its papayas in paper bags.**	Its (no apostrophe) shows ownership: the papayas belong to the fruit stand. It's always means it is.
s3_s0	Set 1: Fix this sentence: Tutu Mahina sold pineapples papayas and mangoes.	**Tutu Mahina sold pineapples, papayas, and mangoes.** (also accepted: Tutu Mahina sold pineapples, papayas and mangoes.)	Use commas to separate the items in a series: pineapples, papayas, and mangoes.
s4_s0	Set 1: Fix this sentence: the baskets on the fruit table is full	**The baskets on the fruit table are full.** (also accepted: The baskets on the fruit table were full.)	Capitalize the first word. The subject is baskets (plural), so use are, not is. The words on the fruit table do not change the subject. End with a period.
s1_s1	Set 2: Fix this sentence: mr. haddad said, the juice tent opens at eight on saturday	**Mr. Haddad said, "The juice tent opens at eight on Saturday."**	Capitalize Mr. Haddad and Saturday (days of the week always start with a capital letter), put quotation marks around his exact words, capitalize the first word of the quote, and put the period inside the closing quotation mark.
s2_s1	Set 2: Fix this sentence: Its warm in Honolulu all year long.	**It's warm in Honolulu all year long.**	It's means it is: it is warm in Honolulu. Use the apostrophe.

ID	Question	Answer	Why (explain to student)
s3_s1	Set 2: Fix this sentence: Caleb weighed the fruit in grams kilograms and pounds.	**Caleb weighed the fruit in grams, kilograms, and pounds.** (also accepted: Caleb weighed the fruit in grams, kilograms and pounds.)	Use commas to separate the items in a series: grams, kilograms, and pounds.
s4_s1	Set 2: Fix this sentence: leilani and caleb carries the fruit boxes to the tent	**Leilani and Caleb carry the fruit boxes to the tent.** (also accepted: Leilani and Caleb carried the fruit boxes to the tent.)	Capitalize the names Leilani and Caleb. Two people (Leilani and Caleb) need the verb carry, not carries. End with a period.
s1_s2	Set 3: Fix this sentence: caleb asked, is 0.06 the same as 0.6	**Caleb asked, "Is 0.06 the same as 0.6?"**	Capitalize Caleb, put quotation marks around his exact words, capitalize the first word of the quote, and put the question mark inside the closing quotation mark.
s2_s2	Set 3: Fix this sentence: Each pineapple has it's own spiky crown.	**Each pineapple has its own spiky crown.**	Its (no apostrophe) shows ownership: the crown (the leafy top) belongs to the pineapple. It's means it is.
s3_s2	Set 3: Fix this sentence: honolulu hilo and kailua are all in hawaii.	**Honolulu, Hilo, and Kailua are all in Hawaii.** (also accepted: Honolulu, Hilo and Kailua are all in Hawaii.; Honolulu, Hilo, and Kailua are all in Hawaiʻi.; Honolulu, Hilo and Kailua are all in Hawaiʻi.; Honolulu, Hilo, and Kailua are all in Hawai'i.; Honolulu, Hilo and Kailua are all in Hawai'i.)	Capitalize the names of cities and states, and use commas to separate the items in a series: Honolulu, Hilo, and Kailua.
s4_s2	Set 3: Fix this sentence: the price of the four jars were $38.00	**The price of the four jars was $38.00.** (also accepted: The price of the four jars is $38.00.)	Capitalize the first word. The subject is price (one thing), not jars, so use was, not were. End with a period.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match 8.4 ÷ 4	**2.1**	2 points if matched on the first try.
match_s0_2	Set 1: Match 0.56 ÷ 7	**0.08**	2 points if matched on the first try.
match_s0_3	Set 1: Match 37.5 ÷ 10	**3.75**	2 points if matched on the first try.
match_s0_4	Set 1: Match 6.2 ÷ 100	**0.062**	2 points if matched on the first try.
match_s0_5	Set 1: Match $9.60 ÷ 3	**$3.20**	2 points if matched on the first try.
match_s1_1	Set 2: Match 9.5 ÷ 5	**1.9**	2 points if matched on the first try.
match_s1_2	Set 2: Match 0.81 ÷ 9	**0.09**	2 points if matched on the first try.
match_s1_3	Set 2: Match 43 ÷ 1,000	**0.043**	2 points if matched on the first try.

ID	Question	Answer	Why (explain to student)
match_s1_4	Set 2: Match 250 centimeters	**2.5 meters**	2 points if matched on the first try.
match_s1_5	Set 2: Match $12.60 ÷ 4	**$3.15**	2 points if matched on the first try.
match_s2_1	Set 3: Match 7.2 ÷ 8	**0.9**	2 points if matched on the first try.
match_s2_2	Set 3: Match 5.1 ÷ 2	**2.55**	2 points if matched on the first try.
match_s2_3	Set 3: Match 68 ÷ 100	**0.68**	2 points if matched on the first try.
match_s2_4	Set 3: Match 1,500 grams	**1.5 kilograms**	2 points if matched on the first try.
match_s2_5	Set 3: Match $14.75 ÷ 5	**$2.95**	2 points if matched on the first try.

BOSS BATTLE: CLOSING TIME AT THE MARKET (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
b1_s0	Set 1: *Leilani and Caleb Have to Plan It All It is the end of a busy Saturday. Mr. Haddad has 30.4 liters of pineapple juice to pour equally into 8 pitchers, and then each pitcher is poured equally into 4 bottles. At the fruit stand, a bag of 6 mangoes costs $9.30, and a big crate of 100 mangoes costs $135.00.* Two steps: how many liters of juice are in each bottle? Liters:	**0.95** (also accepted: 0.95 liters, 0.95 liter, 0.95 L)	Step 1: 30.4 ÷ 8 = 3.8 liters in each pitcher. Step 2: 3.8 ÷ 4 = 0.95 liter in each bottle (3.8 = 3.80, and 380 hundredths ÷ 4 = 95 hundredths). Check: 0.95 × 4 = 3.8 and 3.8 × 8 = 30.4.
b2_s0	Set 1: Three steps: how much MORE does one mango cost in the small bag than in the big crate? Dollars: $	**0.20** (also accepted: $0.20, 0.20 dollars, 0.2 dollars, 20 cents)	Step 1: $9.30 ÷ 6 = $1.55 per mango in the bag. Step 2: $135.00 ÷ 100 = $1.35 per mango in the crate. Step 3: $1.55 − $1.35 = $0.20 more per mango in the bag.
b1_s1	Set 2: *Leilani and Caleb Have to Plan It All It is the end of a busy Saturday. Mr. Haddad's last batch made 45.6 liters of papaya smoothie. He pours it equally into 6 coolers, and then each cooler is poured equally into 10 jars. Later, Leilani and 3 cousins share $26.40 of fruit stand tips equally.* Three steps: how many MILLILITERS of smoothie go in each jar? Milliliters:	**760** (also accepted: 760 milliliters, 760 milliliter, 760 mL)	Step 1: 45.6 ÷ 6 = 7.6 liters in each cooler. Step 2: 7.6 ÷ 10 = 0.76 liter in each jar. Step 3: 0.76 × 1,000 = 760 milliliters.
b2_s1	Set 2: Two steps: after sharing the $26.40 equally, each of the 4 cousins spends $4.75 on shave ice. How much money does each one have left? Dollars: $	**1.85** (also accepted: $1.85, 1.85 dollars)	Leilani and 3 cousins make 4 people. Step 1: $26.40 ÷ 4 = $6.60 each. Step 2: $6.60 − $4.75 = $1.85 left.
b1_s2	Set 3: *Leilani and Caleb Have to Plan It All It is the end of a busy Saturday. Tutu Mahina has 9.6 kilograms of macadamia nuts to pack equally into 4 tubs, and then each tub is packed equally into 100 small bags. Caleb also has a 52.5-meter roll of rope for tying down the tent.* Three steps: how many GRAMS of macadamia nuts go in each small bag? Grams:	**24** (also accepted: 24 grams, 24 gram, 24 g)	Step 1: 9.6 ÷ 4 = 2.4 kilograms in each tub. Step 2: 2.4 ÷ 100 = 0.024 kilogram in each bag. Step 3: 0.024 × 1,000 = 24 grams.

ID	Question	Answer	Why (explain to student)
b2_s2	Set 3: Two steps: Caleb cuts the 52.5-meter roll into 5 equal pieces. Then he trims 0.5 meter of frayed rope off each piece. How many meters long is each piece now? Meters:	**10** (also accepted: 10.0, 10 meters, 10 meter, 10 m)	Step 1: 52.5 ÷ 5 = 10.5 meters for each piece. Step 2: 10.5 − 0.5 = 10 meters.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade4-5/prepost/KCE_Math_PrePost_G4-5W17.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Review: 1/5 ÷ 3 and 4 ÷ 1/3	a) 1/15 b) 12
2	6.4 liters into 2 pitchers	3.2 liters in each pitcher.
3	Equal to 45 ÷ 10	4.5
4	Share $8.60 equally between 2	$4.30 each.
5	Estimate 7.8 ÷ 2	about 4
6	200 centimeters in meters	2 meters

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	14.4 L into 6 pitchers, then each into 10 cups	14.4 ÷ 6 = 2.4 liters per pitcher; 2.4 ÷ 10 = 0.24 liter per cup.
2	Bag of 5 papayas for $9.25 vs. $2.10 each	$9.25 ÷ 5 = $1.85 per papaya in the bag; $2.10 − $1.85 = $0.25 saved on each papaya.
3	1.2 kg into 6 bags, in grams	200 grams (1.2 kg = 1,200 g; 1,200 ÷ 6 = 200).
4	Caleb says 0.56 ÷ 8 = 0.7	He left out the placeholder zero: 56 hundredths ÷ 8 = 7 hundredths, so the quotient is 0.07. Check: 0.07 × 8 = 0.56 (but 0.7 × 8 = 5.6).

★ CHALLENGE

#	Prompt	Model answer / grading note
1	9.6 m into 4 pieces, then each into 10 bows	9.6 ÷ 4 = 2.4 m; 2.4 ÷ 10 = 0.24 m = 24 centimeters per bow.